AW87XXX Android Driver(MTK)

版本： V3.7

时间： 2024-04

contents

1. INFORMATION

Driver File	aw87xxx.c, aw87xxx.h, aw87xxx_device.c, aw87xxx_device.h, aw87xxx_monitor.c, aw87xxx_monitor.h, aw87xxx_dsp.c, aw87xxx_dsp.h, aw87xxx_acf_bin.c, aw87xxx_acf_bin.h, aw87xxx_log.h, aw87xxx_bin_parse.c, aw87xxx_pid_39_reg.h, aw87xxx_pid_59_3x9_reg.h, aw87xxx_pid_59_5x9_reg.h, aw87xxx_pid_5a_reg.h, aw87xxx_pid_18_reg.h, aw87xxx_pid_9b_reg.h, aw87xxx_pid_76_reg.h, aw87xxx_pid_60_reg.h, aw87xxx_pid_c1_reg.h aw87xxx_pid_c2_reg.h
Smart PA	aw87319, aw87329, aw87339, aw87349, aw87359, aw87389, aw87509, aw87519, aw87529, aw87539, aw87549, aw87559, aw87569, aw87579, aw81509, aw87579G aw87390, aw87360, aw87401, aw87390G aw87358, aw87560, aw87550, aw87391,aw87392 aw87565,aw87566,aw81564,aw87567,aw87568,aw87564
I²C Address	aw87358:0x5C others:0x58, 0x59, 0x5A, 0x5B

2. DRIVER PORTING

2.1 AW87XXX Driver Porting

2.1.1 DTS Configuration

AW87359/AW87389/AW87549/AW87390/AW87360/AW87401/AW87390G/AW87391/AW87392 products have no reset pin, so reset GPIO cannot be configured.

Product information sheet

Products	ChipID	reset-pin
AW87319	0x9B	Yes
AW87329/AW87339/AW87349	0x39	Yes
AW87519/AW87529	0x59	Yes
AW87359/AW87389	0x59	**No**
AW87549	0x5A	**No**
AW87559/AW87569/AW87579/AW81509/AW87579G	0x5A	Yes
AW87390/AW87360/AW87401/AW87390G	0x76	**No**
AW87358	0x18	Yes
AW87560/AW87550	0x60	Yes
AW87391/AW87392	0xC1	**No**
AW87565/AW87566/AW81564/AW87567/AW87568/AW87564	0xC2	Yes

Mono PA Configuration

```
&i2c_x {                                          /*x: I2C BUS*/
      aw87xxx_pa@58 {                             /*0x58: I2C Address*/
          compatible = "awinic,aw87xxx_pa";
          reg = <0x58>;
          /* AW87359/AW87389/AW87549/AW87390/AW87360/AW87401/AW87390G/AW87391
products reset GPIO cannot be configured */
          reset-gpio = <&pio 106 0>;  /*Take GPIO 106 as an example */
          dev_index = < 0 >;           /*The first PA configured as 0*/
          /*Set this parameter based on the port id in use*/
          status = "okay";
        };
};
```

Multi PA configuration

```
&i2c_x {
      aw87xxx_pa@58 {
          compatible = "awinic,aw87xxx_pa";
          reg = <0x58>;
          /* AW87359/AW87389/AW87549/AW87390/AW87360/AW87401/AW87390G/AW87391
products reset GPIO cannot be configured */
          reset-gpio = <&pio 106 0>;
          dev_index = < 0 >;            /*The first PA configured as 0*/
          status = "okay";
      };
      aw87xxx_pa@59 {
          compatible = "awinic,aw87xxx_pa";
          reg = <0x59>;
          /* AW87359/AW87389/AW87549/AW87390/AW87360/AW87401/AW87390G/AW87391
products reset GPIO cannot be configured */
          reset-gpio = <&pio 107 0>;
          dev_index = < 1 >;            /* The Second PA configured as 1*/
          status = "okay";
      };
};
    /*The above takes double PA as an example, dev_index of multiple PA increases in
turn. Refer to single PA configuration for other attributes*/
```

2.1.2 Driver Configuration

defconfig Configuration：

```
CONFIG_SND_SOC_AW87XXX=y
```

Create aw87xxx directory in the kernel codecs directory and add driver files:

```
aw87xxx.c,  aw87xxx_device.c,  aw87xxx_monitor.c,  aw87xxx_dsp.c,  aw87xxx_acf_bin.c,
aw87xxx_bin_parse.c  aw87xxx.h,  aw87xxx_device.h,  aw87xxx_monitor.h,  aw87xxx_dsp.h,
aw87xxx_acf_bin.h,  aw87xxx_log.h,  aw87xxx_pid_39_reg.h,  aw87xxx_pid_59_3x9_reg.h,
aw87xxx_pid_59_5x9_reg.h,  aw87xxx_pid_5a_reg.h,  aw87xxx_pid_18_reg.h,
aw87xxx_pid_9b_reg.h,  aw87xxx_pid_76_reg.h,  aw87xxx_pid_60_reg.h,  aw87xxx_pid_c1_reg.h,
aw87xxx_pid_c2_reg.h
```

Kconfig Configuration：

```
config SND_SOC_AW87XXX
    tristate "SoC Audio for awinic AW87XXX Smart K PA"
    depends on I2C
help
    This option enables support for AW87XXX Smart K PA.
```

Makefile Configuration：

```
obj-$(CONFIG_SND_SOC_AW87XXX) +=  aw87xxx/aw87xxx.o
aw87xxx/aw87xxx_device.o  aw87xxx/aw87xxx_monitor.o
aw87xxx/aw87xxx_bin_parse.o  aw87xxx/aw87xxx_dsp.o
aw87xxx/aw87xxx_acf_bin.o
```

2.1.3 BIN Configuration

PA needs to configure register parameters to work normally. The configuration steps of PA bin file are as follows：

Compile Configuration

Add the bin file compilation option at the corresponding location of the platform：

```
PRODUCT_COPY_FILES += \
    xxxx/aw87xxx_acf.bin:$(TARGET_COPY_OUT_VENDOR)/firmware/aw87xxx_acf.bin

/*xxxx means Platform path */
```

Path Configuration

Add the directory of bin file in the firmware_class.c, the general directory is vendor / firmware

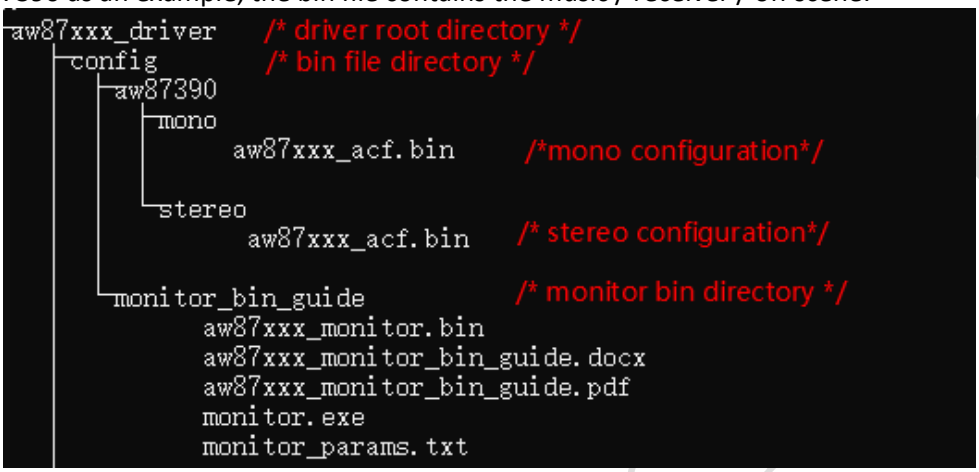
```
    static const char * const fw_path[] = {
    fw_path_para,
        "/vendor/firmware",
        "/lib/firmware/updates/" UTS_RELEASE,
        "/lib/firmware/updates",
        "/lib/firmware/" UTS_RELEASE,
        "/lib/firmware"
    };
```



```
    /* Note: In the debugging stage, you can directly push the bin file to / vendor
/ firmware / */
```

Bin selection

Taking aw87390 as an example, the bin file contains the music / receiver / off scene:



2.2 Platform Driver Configuration

2.2.1 Kcontrol Registration

aw87xxx_codec registration：

```
--- a/mtk-soc-codec-6357.c
+++ b/mtk-soc-codec-6357.c
/*Take the mt6357 platform as an example*/

/*add interface*/
+#ifdef CONFIG_SND_SOC_AW87XXX
+    extern int aw87xxx_add_codec_controls(void *codec);
+#endif

/*add  controls registration */
@@ -5361,6 +5361,15 @@ static int mt6357_codec_probe(struct snd_soc_codec
*codec)
                ARRAY_SIZE(mt6357_pmic_Test_controls));
    snd_soc_add_codec_controls(codec, Audio_snd_auxadc_controls,
                ARRAY_SIZE(Audio_snd_auxadc_controls));
+#ifdef CONFIG_SND_SOC_AW87XXX
+   err = aw87xxx_add_codec_controls((void *)codec);
+   if (err < 0) {
+       pr_err("%s: add_codec_controls failed, err %d\n",
+           __func__, err);
+       return err;
+   };
+#endif
    /* here to set  private data */
    mCodec_data = kzalloc(sizeof(struct mt6357_codec_priv), GFP_KERNEL);
    if (!mCodec_data) {
```

2.2.2 widget Configuration

The configuration of MTK 4G platform is different from that of 5g platform. Please refer to the following:

4G Platform Configuration

Take the mono PA as an example:

```
/* Take the mt6357 platform as an example*/

 /*add interfase and profile definition*/
@@ -80,6 +80,17 @@ static void setDlMtkifSrc(bool enable);
 #ifndef ANALOG_HPTRIM
 static int SetDcCompenSation(bool enable);
 #endif
+#ifdef CONFIG_SND_SOC_AW87XXX
+extern int aw87xxx_set_profile(int dev_index, char *profile);
+
+static char *aw_profile[] = {"Music", "Off"};/*scene in aw87xxx_acf.bin*/
+enum aw87xxx_dev_index {
+      AW_DEV_0 = 0,
+};
+#endif
 static void Voice_Amp_Change(bool enable);

/* add set profile func */
@@ -3296,6 +3347,9 @@ static void Speaker_Amp_Change(bool enable)
       Ana_Set_Reg(AUDDEC_ANA_CON6, 0x0201, 0xffff);
       /* Switch LOL MUX to audio DAC */
       Ana_Set_Reg(AUDDEC_ANA_CON4, 0x011b, 0xffff);
+#ifdef CONFIG_SND_SOC_AW87XXX
+      aw87xxx_set_profile(AW_DEV_0, aw_profile[0]);
+#endif
       /* disable Pull-down HPL/R to AVSS28_AUD */
       if (mIsNeedPullDown)
             hp_pull_down(false);
@@ -3333,6 +3387,10 @@ static void Speaker_Amp_Change(bool enable)
           Ana_Set_Reg(AUDDEC_ANA_CON12, 0x0, 0x1055);
           /* Disable NCP */
           Ana_Set_Reg(AUDNCP_CLKDIV_CON3, 0x1, 0x1);
+
+#ifdef CONFIG_SND_SOC_AW87XXX
+      aw87xxx_set_profile(AW_DEV_0, aw_profile[1]);
+#endif
           TurnOffDacPower();
       }
   }
```

5G Platform Configuration

Take the mono PA as an example:

```
/* Take the mt6357 platform as an example*/
```



```
/*add interfase and profile definition*/
@@ -17,6 +17,13 @@
 #include "../../codecs/mt6359.h"
 #include "../common/mtk-sp-spk-amp.h"

+#ifdef CONFIG_SND_SOC_AW87XXX
+extern int aw87xxx_set_profile(int dev_index, char *profile);
+static char *aw_profile[] = {"Music", "Off"};
+enum aw87xxx_dev_index {
+     AW_DEV_0 = 0,
+};
+#endif

/* add set profile func */
 /*
  * if need additional control for the ext spk amp that is connected
@@ -92,9 +99,25 @@ static int mt6853_mt6359_spk_amp_event(s
    switch (event) {
    case SND_SOC_DAPM_POST_PMU:
        /* spk amp on control */
+#ifdef CONFIG_SND_SOC_AW87XXX
+       ret = aw87xxx_set_profile(AW_DEV_0, aw_profile[0]);
+       if (ret < 0) {
+           pr_err("[Awinic] %s: set profile[%s] failed",
+                  __func__, aw_profile[0]);
+           return ret;
+       }
+#endif
        break;
    case SND_SOC_DAPM_PRE_PMD:
        /* spk amp off control */
+#ifdef CONFIG_SND_SOC_AW87XXX
+       ret = aw87xxx_set_profile(AW_DEV_0, aw_profile[1]);
+       if (ret < 0) {
+           pr_err("[Awinic] %s: set profile[%s] failed",
+                  __func__, aw_profile[1]);
+           return ret;
+       }
+#endif
        break;
    default:
        break;
```

2.3 Low Battery Protection Function

2.3.1 DSP Communication Interface

If the platform is equipped with DSP, please go to open MTK platform macro definition in aw_dsp.h

```
#define AW_MTK_OPEN_DSP_PLATFORM
```


2.4 Drive Verification

Take aw87560 single PA registration as an example：

1）**I2C communication successful：**

```
[Awinic] aw87xxx_pa_init:driver version: Dv2.4.0.5
[Awinic] [0-0058]aw87xxx_malloc_init: struct aw87xxx devm_kzalloc and init down
[Awinic] [0-0058]aw87xxx_dtsi_parse: parse dev_index=[0]
[Awinic] [0-0058]aw87xxx_dtsi_parse: reset gpio[1160] parse succeed
[Awinic] [0-0058]aw87xxx_device_parse_port_id_dt: rx-port-id: 0xb030
[Awinic] [0-0058]aw87xxx_device_parse_topo_id_dt: rx-topo-id: 0x1000ff01
[Awinic] [0-0058]aw87xxx_dev_hw_pwr_ctrl: hw power on
[Awinic] [0-0058]aw87xxx_dev_get_chipid: read chipid[0x60] succeed
[Awinic] [0-0058]aw_dev_chip_init: product is pid_60 class
```

2）**PA Bin loaded successfully：**

```
[Awinic] [0-0058]aw_get_prof_count_v_0_0_0_1: get profile count=[3]
[Awinic] [0-0058]aw_get_vaild_prof_v_0_0_0_1: product=[aw87560]----profile=[Music]
[Awinic] [0-0058]aw_get_vaild_prof_v_0_0_0_1: product=[aw87560]----profile=[Receiver]
[Awinic] [0-0058]aw_set_prof_off_info_v_0_0_0_1: product=[aw87560]----profile=[Off]
[Awinic] [0-0058]aw_get_vaild_prof_v_0_0_0_1: get vaild profile succeed
[Awinic] [0-0058]aw_set_prof_name_list_v_0_0_0_1: index=[0], profile_name=[Music]
[Awinic] [0-0058]aw_set_prof_name_list_v_0_0_0_1: index=[1], profile_name=[Receiver]
[Awinic] [0-0058]aw_set_prof_name_list_v_0_0_0_1: index=[2], profile_name=[Off]
[Awinic] [0-0058]aw_parse_acf_v_0_0_0_1: acf parse success
[Awinic] [0-0058]aw87xxx_init_default_prof: init profile name [Off]
[Awinic] [0-0058]aw87xxx_fw_load: acf parse succeed
[Awinic] [0-0059]aw87xxx_fw_load: enter
```

3）**Kcontrol registration successful：**

```
[Awinic] [0-0058]aw87xxx_kcontrol_dynamic_create: enter
[Awinic] [0-0058]aw87xxx_kcontrol_dynamic_create: add codec controls[aw87xxx_profile_switch_0,aw87xxx_vmax_get_0,aw87xxx_monitor_switch_0]
[Awinic] [0-0058]aw87xxx_public_kcontrol_create: enter
[Awinic] [0-0058]aw87xxx_public_kcontrol_create: add public codec controls[aw87xxx_spin_switch]
[Awinic] [0-0058]aw87xxx_profile_switch_get: current profile:[Off]
```

4）**Set profile successful：**

```
kona:/sys/bus/i2c/drivers/aw87xxx_pa/0-0058 # cat profile
 Music
 Receiver
>Off
kona:/sys/bus/i2c/drivers/aw87xxx_pa/0-0058 # echo Music > profile
[Awinic] [0-0058]aw87xxx_attr_get_profile: current profile:[Off]
[Awinic] [0-0058]aw87xxx_attr_set_profile: set profile [Music]
[Awinic] [0-0058]aw87xxx_update_profile: load profile[Music] enter
[Awinic] [0-0058]aw87xxx_monitor_stop: enter
[Awinic] [0-0058]aw87xxx_power_on: enter
[Awinic] [0-0058]aw87xxx_acf_get_prof_desc_form_name: enter
[Awinic] [0-0058]aw87xxx_acf_get_prof_desc_form_name: get prof desc down
[Awinic] [0-0058]aw87xxx_power_on: get profile[Music] data len [84]
```

5）**Play music successful：**

```
[Awinic] aw87xxx_set_profile_by_id:aw87xxx, dev_index[0] set profile[Music] by id[1]
[Awinic] [0-0058]aw87xxx_set_profile: set dev_index = 0, profile = Music
[Awinic] [0-0058]aw87xxx_update_profile: load profile[Music] enter
[Awinic] [0-0058]aw87xxx_monitor_stop: enter
[Awinic] [0-0058]aw87xxx_power_on: enter
[Awinic] [0-0058]aw87xxx_acf_get_prof_desc_form_name: enter
[Awinic] [0-0058]aw87xxx_acf_get_prof_desc_form_name: get prof desc down
[Awinic] [0-0058]aw87xxx_power_on: get profile[Music] data len [84]
[Awinic] [0-0058]aw87xxx_dev_hw_pwr_ctrl: hw power on
[Awinic] [0-0058]aw87xxx_dev_reg_update: reg=0x76, val = 0x92
[Awinic] [0-0058]aw87xxx_dev_reg_update: reg=0x7d, val = 0x7a
[Awinic] [0-0058]aw87xxx_dev_reg_update: reg=0x7c, val = 0xc1
```

3. DEBUG INTERFACE

3.1 Device node

AW87XXX Driver will create multiple device node files with different functions. The path is sys/bus/i2c/driver/aw87xxx_smartpa/*-00xx, where * is the i2c bus number and xx is the i2c address.

reg

Node name	reg
Description	read and write all register value of aw87xxx
Instruction	read register value：echo reg_addr > rw (Hexadecimal operation) cat rw write register value：echo reg_addr reg_data > rw (Hexadecimal operation)
Example	cat reg （get all the values of the register with read permission） echo 0x01 0x07 > reg（write the value of 0x07 to the register of 0x01）

profile

Node name	profile
Description	scene switch
Instruction	View the current scene and switchable scenes：cat profile Select scene：echo xxxx > profile (xxxx means scene name)
Example	cat profile echo “Music” > profile （Select Music scene） echo “Off” > profile （Select Off scene）

hwen

Node name	hwen
Description	Control PA status
Instruction	cat hwen （get PA status） echo 1 > hwen （aw87xxx power on） echo 0 > hwen （aw87xxx power down）

esd_enable

Node name	esd_enable
Description	Esd function switch
Instruction	cat esd_enable （Get status of ESD function） echo true > esd_enable （Enable the ESD function） echo false > esd_enable （Disable the ESD function）

vmax

Node name	vmax
Description	set vmax to dsp and get the current vmax value of dsp
Instruction	cat vmax （Get the value of the current vmax） echo N > vmax （Send calculated vmax value）
Example	echo 0xfff95f7e > vmax （Send 0xfff95f7e value to dsp）

vbat

Node name	vbat
Description	set the current power and get the real-time power value
Instruction	cat vbat （Get real-time value） echo capacity > vbat （Set vbat） echo 0 > vbat （Cancel input vbat before）
Example	echo 50 > vbat （Set the current power to 50%）

monitor

Node name	monitor
Description	Enable or disable monitor function
Instruction	cat monitor （get monitor switch status） echo 0 > monitor （disable monitor） echo 1 > monitor （enable monitor）

monitor_count

Node name	monitor_count
Description	Set the monitor interval for Vmax
Instruction	cat monitor_count （get monitor interval） echo 5 > monitor_count （Set the monitor interval to 5）

monitor_time

Node name	monitor_time
Description	Set monitor interval time(ms)
Instruction	cat monitor_time （get monitor interval time） echo N > monitor_time （set monitor interval time）

rx

Node name	rx
Description	Set or get dsp status
Instruction	cat rx （get rx status of dsp） echo 1 > rx （enable dsp rx module） echo 0 > rx （disable dsp rx module）

monitor_update

Node name	monitor_update
Description	update monitor parameter configuration
Instruction	echo 1 > monitor_update （update monitor parameter）

4. APPENDIX

4.1 FAQ

4.1.1 White noise appears after transplanting the AWINIC algorithm

If there is regular white noise playback during PA operation (audio source 10s ->white noise 3s->audio source 10s ->white noise 3s->……)

The above phenomenon indicates that the awinic algorithm authentication has failed. Please enable the corresponding function in the driver to recompile.

```
    diff --git a/aw87xxx_device.h b/aw87xxx_device.h
index 6234a73..ad62b2c 100644
--- a/aw87xxx_device.h
+++ b/aw87xxx_device.h
@@ -64,7 +64,7 @@
  ****************************************/
 struct aw_device;

-/*#define AW_ALGO_AUTH_DSP*/
+#define AW_ALGO_AUTH_DSP
 extern int g_algo_auth_st;

     enum AW_ALGO_AUTH_MODE {
```